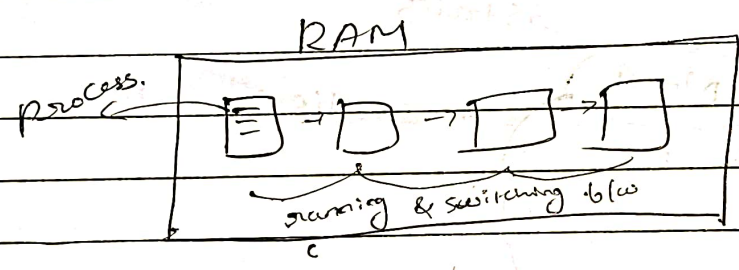
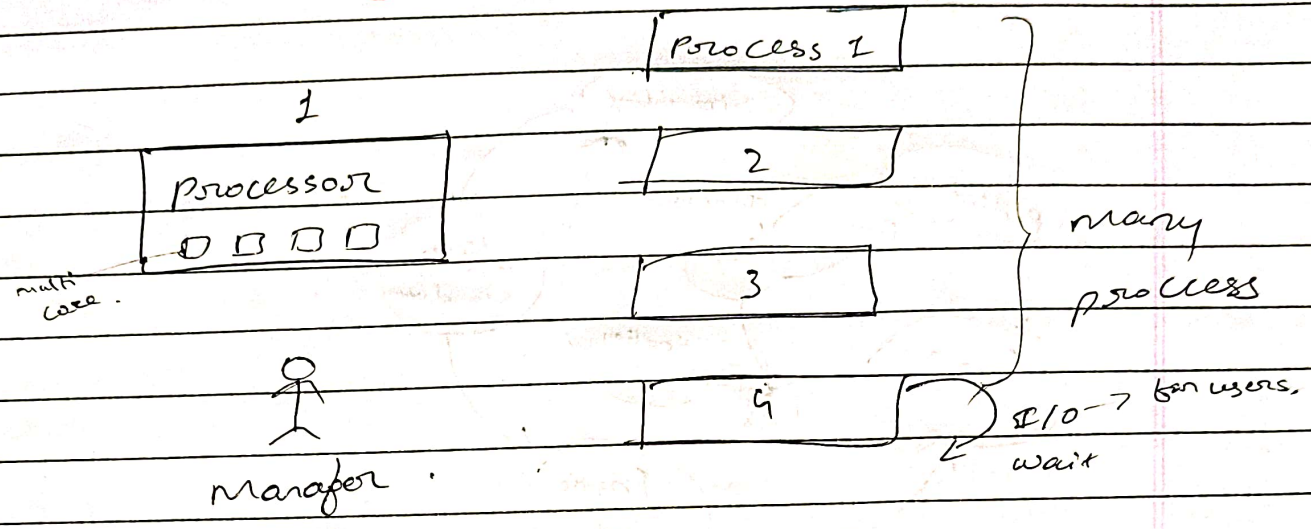
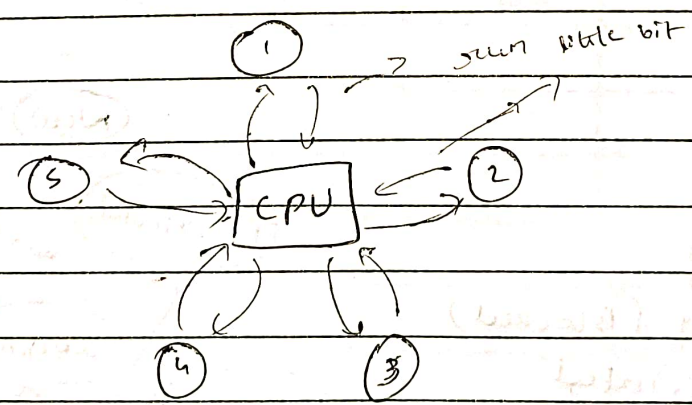


1/3 Process :

* Process Management & Threads

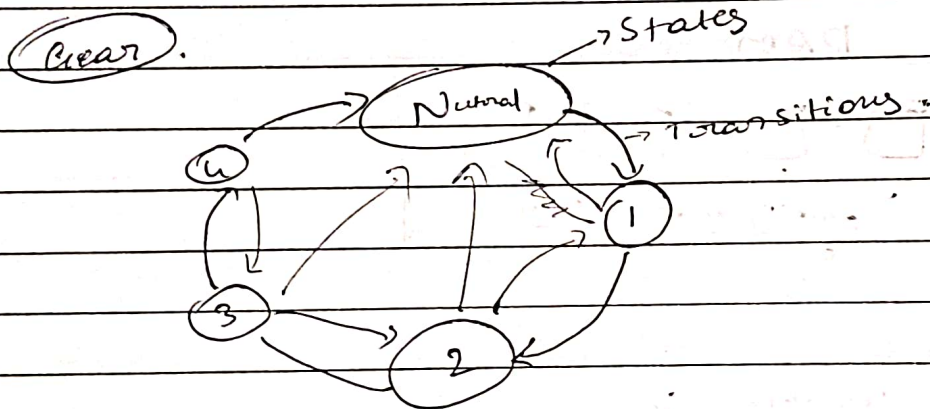
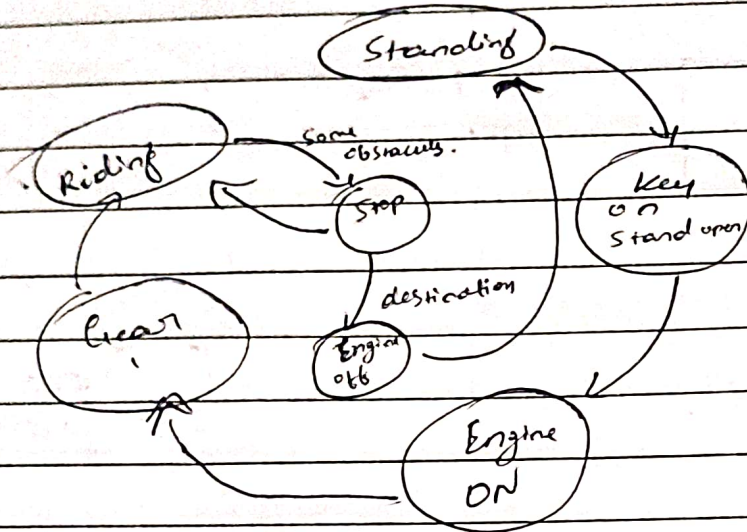


* Round Robin :



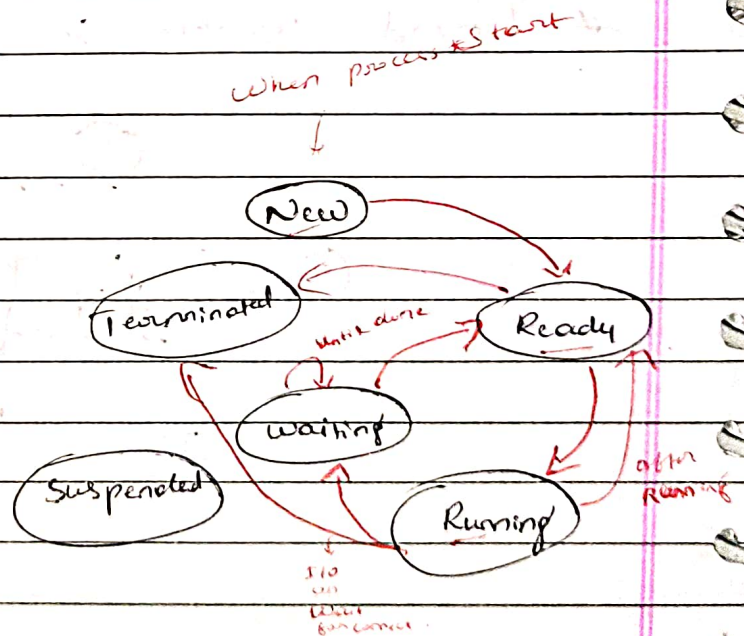
Process State :

Ex :- bike :-

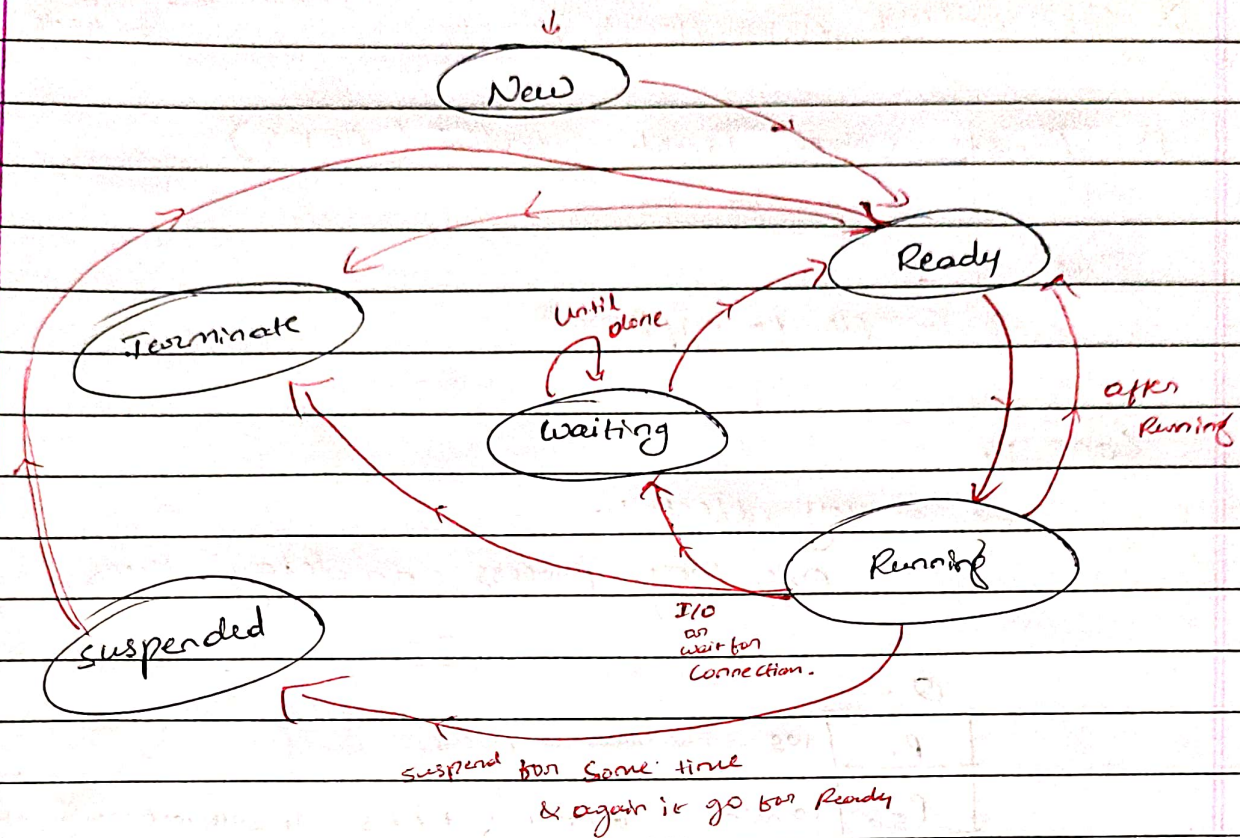


States :

- 1) New
- 2) Ready
- 3) Running
- 4) Waiting (Blocked)
- 5) Terminated
- 6) Suspended.



When process starts in RAM.

Process table :-

ID	Name	loc ²	Status
-	-	-	Running
-	-	-	Ready
-	-	-	
-	-	-	
-	-	-	

Scheduling Algorithms :-

1) First-Come, First-Served (FCFS)

Pros :-

→ Simple to implement

Cons :-

→ Convoy effect :-

one long process can delay others.

Q.

P ₁	10s
P ₅₀	10 min
P ₁₀₀	10s

→ time takes to process.

because of FCFS it will take 10 min time

& others are wait

wait for 10 min.

↓
this is called

(process starvation)

2) Shortest Job Next (SJN) / Shortest Job First (SJF)

pros:-

minimizes avg waiting time.

cons :-

→ hard to predict burst time

→ may cause starvation.

process	Time
P ₁	10 sec
P ₂	25s
P ₃	4
P ₆	3
P ₇	2
P ₈	10

→ 3rd

but it is 4th but when it start another small time process comes that's

→ 2nd

→ 1st execute

→ 4th

→ 5th

way it wait this is conv

↓
Starvation

3) Priority Scheduling

Pros :-

→ Important tasks get more CPU time.

Cons :-

→ Starvation of low-priority process.

Solⁿ :- Aging technique (gradually increasing priority).

Process	Priority
Virus	P_0
Display	P_0
Email	P_1
Spotify	P_2
Download	P_3

high priority

lowest priority

→ Starvation of low priority (when high priority comes more)

4) Round Robin (RR)

Descripⁿ :- Each process gets a fixed time slice

Pros :-

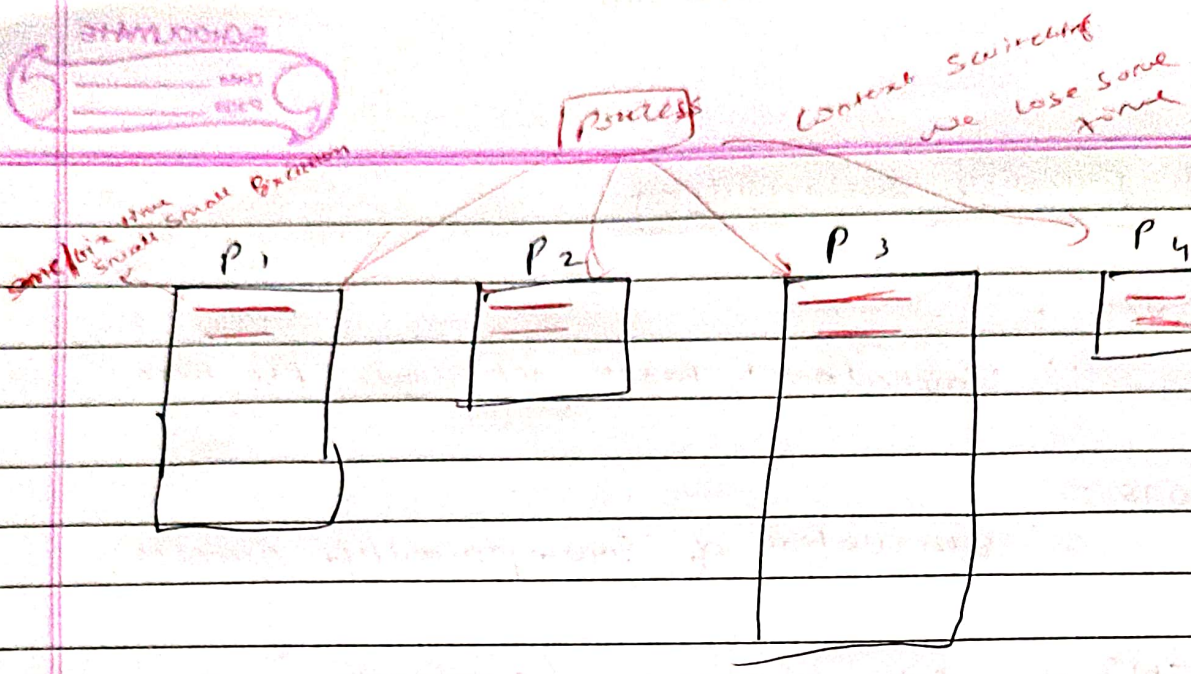
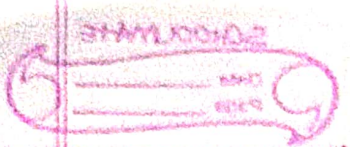
→ Fair & responsive

Cons :-

→ Overhead of frequent context switches.

Use case :- Time-Sharing Systems.

↓ lose some time



5) multilevel Queue Scheduling

(simple queue paid, force cons ↓ force not jumped to paid)

→ processes are divided into queues.
(ex: interactive, batch).

pros:-

Special handling for different types of processes

cons:-

Rigid queue boundaries.

6) multilevel Feedback Queue:-

(simple → It jumps for paid queue. emergency/need)

→ Processes can move b/w queues based on behavior.

pros:-

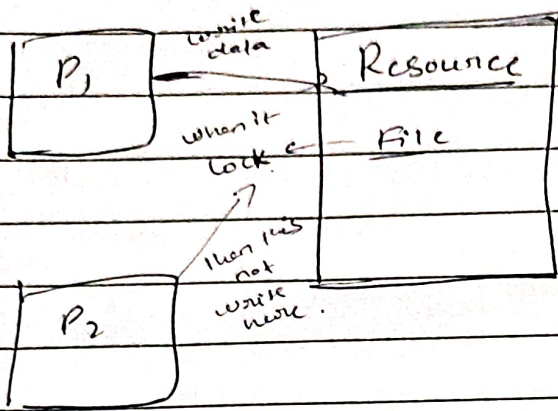
→ more flexible than multilevel queue

Cons:-

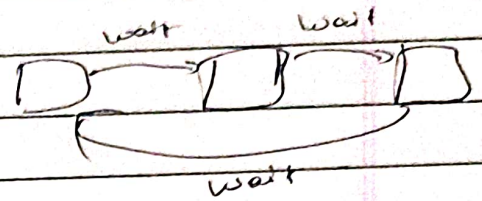
Complex to implement

* Dead lock :-

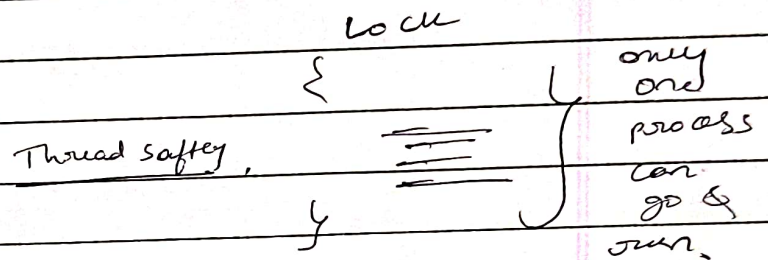
Threads
 not safe → Dead lock.



Dead lock :-



Critical Situation



defⁿ :- Two or more processes waiting on each other indefinitely.

* Blue Screen of Death (BSOD) :-

→ Occurs in Windows when kernel encounters critical error.

→ Common causes :-

- Driver faults
- Hardware failures
- Invalid memory access.

→ dumps memory content for debugging.